Comparing Generalization in Learning with Limited Numbers of Exemplars: Transformer vs. RNN in Attractor Dynamics

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ABSTRACT

ChatGPT, a widely-recognized large language model (LLM), has recently gained substantial attention for its performance scaling, attributed to the billions of web-sourced natural language sentences used for training. Its underlying architecture, Transformer, has found applications across diverse fields, including video, audio signals, and robotic movement. However, this raises a crucial question about Transformer's generalization in learning (GIL) capacity. Is ChatGPT's success chiefly due to the vast dataset used for training, or is there more to the story? To investigate this, we compared Transformer's GIL capabilities with those of a traditional Recurrent Neural Network (RNN) in tasks involving attractor dynamics learning. For performance evaluation, the Dynamic Time Warping (DTW) method has been employed. Our simulation results suggest

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that under conditions of limited data availability, Transformer's GIL abilities are markedly inferior to those of RNN.

INTRODUCTION

Among human competencies, the ability to extract generalized knowledge or skills from a relatively limited set of experiences is particularly fascinating [1, 2, 3]. Enhancing this ability, which is known as generalization in learning (GIL), in machines has been a major objective in machine learning and AI [4, 5]. The crux of GIL lies in extracting the essential structure underlying a dataset, while disregarding contaminating noise, rather than memorizing each data point individually.

Deep learning, with its impressive scalability, has recently attracted great attention [6, 7]. The capability of Transformer [8], ChatGPT's core engine, to correctly answer questions across a diverse array of knowledge domains far surpasses that of the average human. However, a key concern is that Transformer's remarkable performance is reliant on billions of training sentences—a significant divergence from human learning, which requires far fewer experiences to develop knowledge or skills [7, 9]. Against this background, we propose a hypothesis: while the Transformer excels in scaling when fed vast amounts of data, its generalization capability is considerably weaker than those of other neural network models, especially the Recurrent Neural Network (RNN), when working with a small number of exemplars.

Previous research has undertaken comparative evaluations of performance between Transformer and RNN architectures across various tasks. For instance, Tran et al. [10] focused on tasks involving hierarchical structure modeling, such as subject-verb agreement [11], and emphasized the importance of recurrency. Dehghani et al. [12] employed the bAbi question-answering dataset [13] and

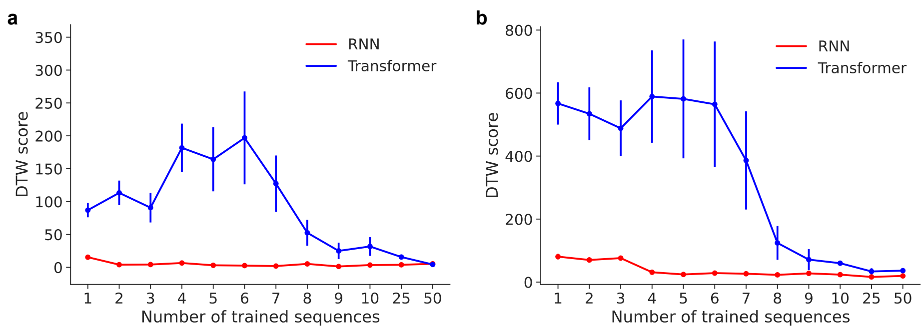

Figure 1: **Quantitative evaluation of model performance using Dynamic Time Warping (DTW). a, b,** The figure presents the average DTW plot along with the standard error for models trained with point and circle attractors, respectively.

suggested the efficacy of incorporating an adequate recurrent inductive bias for several algorithmic and language understanding tasks. While these studies employed relatively simple tasks, the number of training samples employed in these studies still surpasses the number required for human learning, which requires a few experiences to develop knowledge and skills. Therefore, in contrast to these prior works, we provide a simple attractor dynamics learning task to compare GIL in extracting essential structure from a limited set of exemplars. Importantly, our approach focuses on the generative model and involves training the Transformer model without extensive datasets, allowing us to assess their inherent generalization capabilities.

RESULTS

The ability to extract essential structures underlying a limited number of experiences is a core AI capability. This experiment was designed to explore the capacity of generalization in learning (GIL) in both RNN and Transformer models. Models were trained using two forms of attractor dynamics, i.e., point and